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Aero 4630

Homework 2

1.

- a. Find the general solution of the following homogenous ODE for a spring-damper system where $F_{ext} = 0$ and no initial conditions are prescribed

$$\dot{x}(t) = -3x(t)$$

$$\frac{dx}{dt} = -3x$$

$$\frac{dx}{-3x} = dt$$

$$-\frac{1}{3} \ln(-3x) = t + C_1$$

$$-3x = e^{-3t+3C_1}$$

$$x = -\frac{C_2}{3} e^{-3t}$$

- b. Consider the same system with a constant external force $F_{ext} = 1$, Obtain the general solution and verify it by plugging it back to the ODE. Provide a physical explanation of the solution.

$$\dot{x}(t) = -3x(t) + 1$$

$$\frac{dx}{dt} = -3x + 1$$

$$\frac{dx}{-3x + 1} = dt$$

$$-\frac{1}{3} \ln(-3x + 1) = t + C_1$$

$$-3x + 1 = e^{-3t+3C_1}$$

$$x = \frac{e^{-3t+3C_1} - 1}{-3}$$

$$x = \frac{C_2 e^{-3t} - 1}{-3}$$

$$\dot{x} = C_2 e^{-3t}$$

Check : $C_2 e^{-3t} = -3 \left(\frac{C_2 e^{-3t} - 1}{-3} \right) + 1 = C_2 e^{-3t} - 1 + 1 = C_2 e^{-3t}$, valid

At infinity, the spring comes to rest at $x=1/3$, which mean's its k value is 3. It should decrease slower than a's ODE at a given starting position.

- c. Now, consider the same system, but the external force $F_{ext} = \sin(t)$, which varies in time. Obtain its general solution and verify it by plugging it back into the ODE. Provide a physical explanation of the solution

$$\dot{x}(t) = -3x(t) + \sin(t)$$

$$\dot{x}(t) + 3x(t) = \sin(t)$$

Undetermined coefficients

$$x_p = A\sin(t) + B\cos(t)$$

$$A\cos(t) - B\sin(t) + 3A\sin(t) + 3B\cos(t) = \sin(t)$$

$$3A - B = 1$$

$$A + 3B = 0$$

$$-9B - B = 1, B = -\frac{1}{10}$$

$$A = \frac{3}{10}$$

$$x_p = \frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)$$

$$x = Ce^{-3t} + \frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)$$

Check

$$\dot{x}(t) + 3x(t) = \sin(t)$$

$$-3Ce^{-3t} + \frac{3}{10}\cos(t) + \frac{1}{10}\sin(t) + 3Ce^{-3t} + \frac{9}{10}\sin(t) - \frac{3}{10}\cos(t) = \sin(t)$$

$$\frac{3}{10}\cos(t) + \frac{1}{10}\sin(t) + \frac{9}{10}\sin(t) - \frac{3}{10}\cos(t) = \sin(t)$$

$$\frac{1}{10}\sin(t) + \frac{9}{10}\sin(t) = \sin(t)$$

$$\frac{10}{10}\sin(t) = \sin(t)$$

The equation should start out larger, then decrease in magnitude as t approaches infinity because of Ce^{-3t} , where it will settle on whatever sinusoid is created by the $\frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)$ terms.

- d. For the same system as in part c, where $F_{\text{ext}} = \sin(t)$ consider the initial condition of $x(0) = 0.3$. obtain the solution to the ode and verify it does satisfy the initial condition.

$$x = Ce^{-3t} + \frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)$$

$$0.3 = C + 0 - \frac{1}{10}$$

$$C = 0.4$$

$$x = \frac{4}{10}e^{-3t} + \frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)$$

$$0.3 = \frac{4}{10} + 0 - \frac{1}{10} = \frac{3}{10} = 0.3$$

- e. Find the solution to the following ode without directly solving it. Use the principle of superposition and the solutions you already have for parts b and c. verify your solution by plugging it back into the ODE and checking the boundary condition.

$$\dot{x}(t) = -3x(t) + 2\sin(t) + 6, x(0) = 0$$

$$x = \frac{C_2 e^{-3t} - 1}{-3}$$

$$x = Ce^{-3t} + \frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)$$

Superposition answer:

$$x = Ce^{-3t} + 2\left(\frac{3}{10}\sin(t) - \frac{1}{10}\cos(t)\right) + 6\left(\frac{1}{3}\right)$$

$$x = Ce^{-3t} + \frac{3}{5}\sin(t) - \frac{1}{5}\cos(t) + 2$$

$$0 = C - \frac{1}{5} + 2$$

$$C = -\frac{9}{5}$$

$$x = -\frac{9}{5}e^{-3t} + \frac{3}{5}\sin(t) - \frac{1}{5}\cos(t) + 2$$

Check

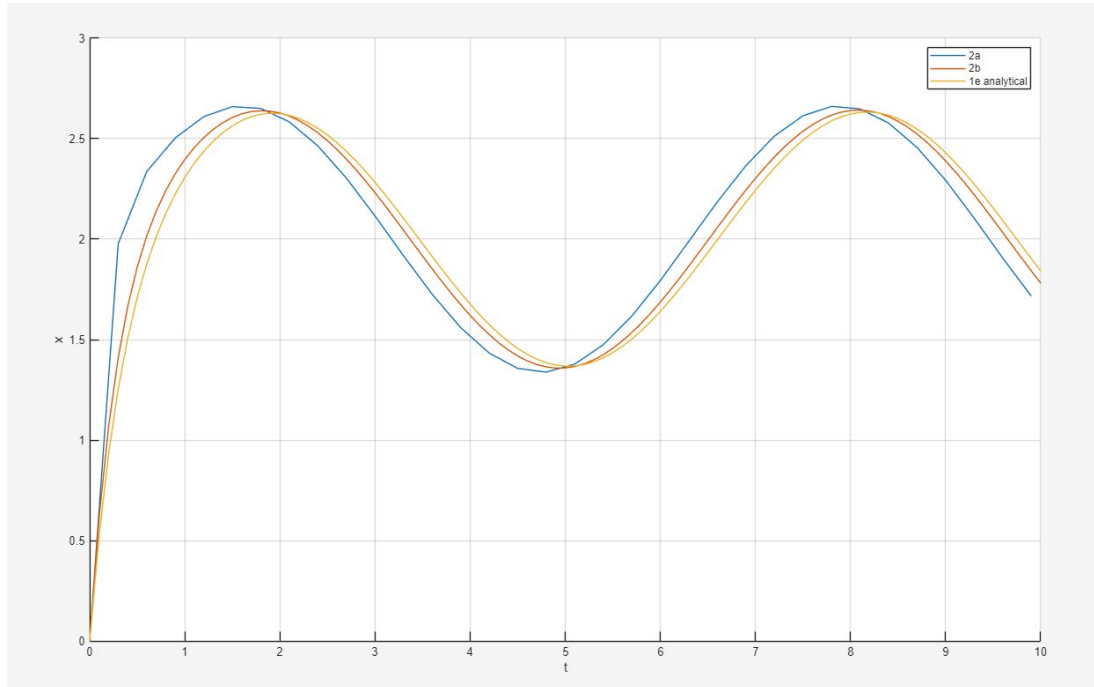
$$\frac{27}{5}e^{-3t} + \frac{3}{5}\cos(t) + \frac{1}{5}\sin(t) = \frac{27}{5}e^{-3t} - \frac{9}{5}\sin(t) + \frac{3}{5}\cos(t) - 6 + 2\sin(t) + 6$$

$$\frac{1}{5}\sin(t) = -\frac{9}{5}\sin(t) + 2\sin(t) = \frac{1}{5}\sin(t)$$

$$0 = -\frac{9}{5} + 0 - \frac{1}{5} + 2 = 0$$

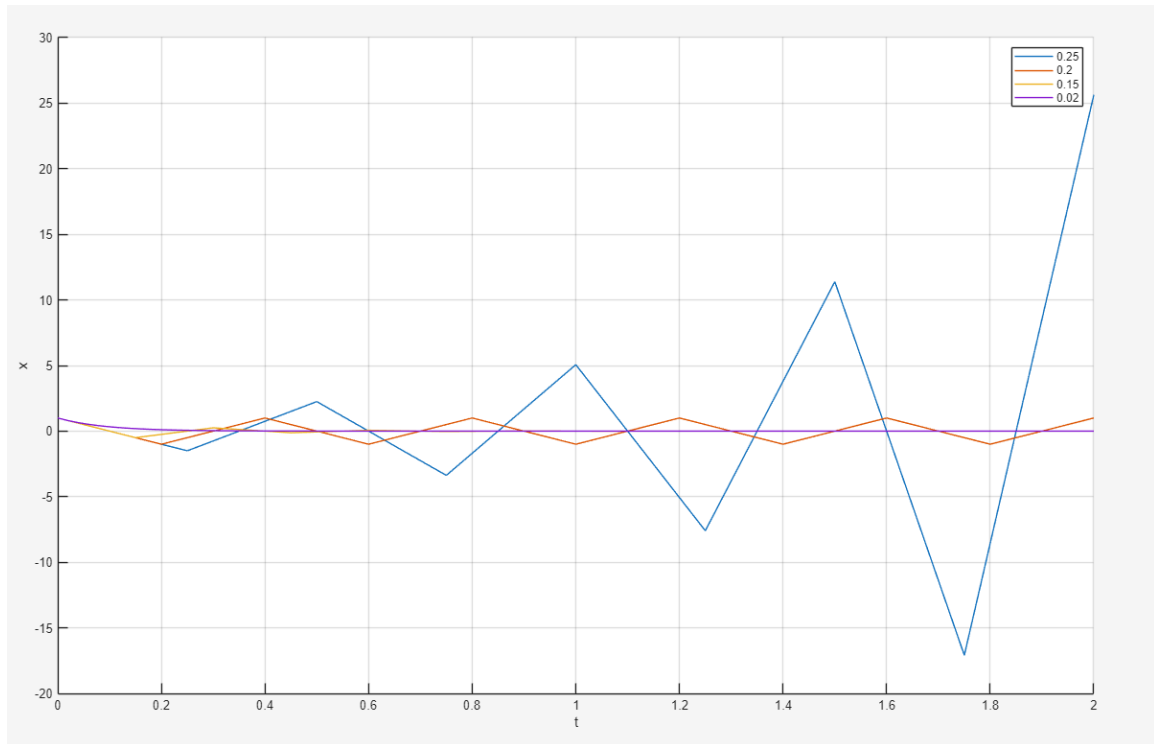
2.

- a. Use Euler integration $\Delta t = 0.3$ in Matlab to approximate the solution to problem 1e from $0 < t < 10$.
- b. Use Euler integration $\Delta t = 0.1$ in Matlab to approximate the solution to problem 1e from $0 < t < 10$.
- c. Plot the analytical solution from 2(e), Euler integration approximation from part 2(a) and 2(b) in a single plot in Matlab. Compare these three solutions and discuss the results you have.



Reducing the timestep made the Euler integration approach the analytical solution to the problem, although there is still visible error on the 0 to 10 time scale. I am glad to see they fit pretty close though as it gives me confidence in my analytical solution.

- d. Use Euler integration to approximate the solution to the ODE. Compare the results and discuss your observations.



The most glaring observation is that the largest timestep, instead of converging like the rest of the integrations actually diverged because of its error, besides that, 0.15 and 0.02 both converged to 0 while it appears that 0.2 might oscillate around 0 as t approaches infinity.

3. Consider a vector $\mathbf{v}$ in the xy -plane with coordinates (x, y) . We wish to rotate this vector counter-clockwise by an angle θ about the origin to obtain a new vector $\mathbf{v}'$ with coordinates (x', y')
- a. Let the standard basis vectors be $\mathbf{e}_1$ and $\mathbf{e}_2$. After rotation by θ , these become $\mathbf{e}_1'$ and $\mathbf{e}_2'$. Using trigonometry, find the coordinates of $\mathbf{e}_1'$. Similarly, find the coordinates of $\mathbf{e}_2'$.

$$\mathbf{e}_1' = \begin{bmatrix} \cos \theta \\ \sin(\theta) \end{bmatrix}; \mathbf{e}_2' = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}$$

For $\theta = 0$

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- b. Any vector $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ can be written as a linear combination of the original basis vectors: $\mathbf{v} = x\mathbf{e}_1 + y\mathbf{e}_2$. Assuming rotation is a linear transformation, the rotated vector is:

$$\mathbf{v}' = x\mathbf{e}_1' + y\mathbf{e}_2'$$

Substitute your expressions for $\mathbf{e}_1'$ and $\mathbf{e}_2'$ from part a into this equation. Write out the resulting expression for x' and y' in terms of x, y and θ .

$$\mathbf{v}' = x(\mathbf{e}_1 \cos(\theta) + \mathbf{e}_2 \sin(\theta)) + y(-\mathbf{e}_1 \sin(\theta) + \mathbf{e}_2 \cos(\theta))$$

$$x' = x \cos(\theta) + y \sin(\theta)$$

$$y' = -x \sin(\theta) + y \cos(\theta)$$

- c. Observe that your result can be written as:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Fill in the entries of the 2×2 matrix using trigonometric functions of θ . This is the rotation matrix.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$